

Medical Image Computing

A Long Technical Handbook — Summer of Code mentee guide

Mentor's handbook (v1)

How to read this chapter

This handbook is your **big-picture-first** reference. Each weekly chapter sketches every term you will encounter that week *in moderately technical language*, with hyperlinks to longer reads. The pattern of reading you should adopt every week is:

1. **Read this chapter end-to-end.** Roughly 15–30 minutes. Goal: when you next hear “Cauchy–Riemann” or “Hammersley–Clifford” or “ELBO”, you can locate it on your mental map.
2. **Open the week’s slide deck** (see `slides/SLIDES_INDEX.md`). Read it linearly; you will recognise the terms.
3. **Pick at most two** linked resources per week and read them properly. The rest are there if you decide to specialise later.
4. **Do the notebooks and exercises** in `week_N/Readme.md`.

Why this order? Because most students lose more time hopping between random search results trying to figure out what a term means than they do actually learning the concept. The handbook exists to give you a *scaffold* so that every external resource you read already has a slot to drop into.

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1 Course philosophy: math first, then DL

Medical Image Computing (MIC) is the engineering and mathematical discipline that builds algorithms for medical-image data. It sits at the intersection of computer science, applied mathematics, signal / image processing, statistics, and the physics of imaging modalities (X-rays, magnetic resonance, ultrasound, microscopy, ...).

This course is built on top of *CS-736 Medical Image Computing* at IIT Bombay (Prof. Suyash P. Awate). Two emphases follow from that:

- It is **math-first**. Every algorithm in weeks 2–6 is either *a maximum-a-posteriori (MAP) estimate under some prior and noise model*, or *a clustering / classification / regression problem with a chosen geometry*. Deep learning is a powerful *parameterisation*, but it does not replace the principled scaffold.
- It is **modality-aware**. CT and MRI images are produced by physical systems with very specific structure; the algorithms that work on them inherit that structure (Beer–Lambert law, Bloch equations, Radon transform, k -space).

The recurring object in everything we do is Bayes’ theorem applied to an unknown image (or label / shape / mapping) x given measurements y :

$$p(x | y) \propto \underbrace{p(y | x)}_{\text{noise / forward model}} \cdot \underbrace{p(x)}_{\text{prior}}. \quad (1)$$

Almost everything you learn this summer is one of:

- a way to write down $p(y | x)$ (Weeks 2, 3);
- a way to write down $p(x)$ (Weeks 2, 4, 5, 6);
- a way to maximise the right-hand side of (1) (Weeks 3, 5);
- a way to evaluate whether your $\hat{x}$ is any good (Weeks 5, 7, 8).

2 Week 1 — Images, math & probability foundations

2.1 Big picture

A **digital image** is just an N -dimensional array of numbers laid out on a regular grid. A pixel (2D) or voxel (3D) is one entry in that array; its *intensity* is the number stored there. What that number physically means depends on the modality: for CT it is an attenuation coefficient (or its Hounsfield-unit rescaling); for MRI it is the magnitude of a complex magnetisation; for microscopy it is a photon count.

The colour you *see* when an image is displayed is not the data — it is a **colormap**, a function from the data values to colours. The same MRI slice can look black-on-white or white-on-black; the data are identical. Get into the habit of always inspecting the *intensity histogram* of an image alongside the picture.

Two pieces of mathematical machinery sit underneath nearly every algorithm in this course:

- **Linear algebra** — because images are vectors (just very long ones), and almost every operation we do is a linear map followed by something. Make sure you are comfortable with: matrix–vector multiplication as a linear map, transposes, inverses, ranks, eigen-decomposition of symmetric matrices, the singular value decomposition (SVD) at the level of “it factorises $A = U\Sigma V^\top$ and the columns of V are the input principal directions”.
- **Probability & statistics** — because every imaging system produces noisy measurements, and every prior we impose on x is a probability distribution. You should be able to write the PDF of a Gaussian and a Poisson by hand, and state Bayes’ theorem without looking it up.

2.2 Vocabulary you will meet

Pixel / voxel. A single sampling location of the image grid. “Pix-el” = *picture element*. Voxel is the 3D analogue.

Intensity. The value stored at a pixel. Real or integer; signed or unsigned; range depends on bit-depth ($[0, 255]$ for 8-bit, $[0, 65535]$ for 16-bit, real for float scanners).

Colormap. A function from intensity to displayed colour. Examples: `gray`, `viridis`, `hot-iron`.

Voxel spacing. The physical size (in mm) of one voxel along each axis. Medical scanners produce *anisotropic* voxels (e.g. $1 \times 1 \times 3$ mm) for many protocols.

Convolution. The operation $(I * k)[m, n] = \sum_{i,j} I[m - i, n - j] k[i, j]$. Equivalent in spectrum: pointwise product in the Fourier domain.

2D Fourier transform. Decomposes an image into a sum of plane waves indexed by spatial frequency (u, v) .

Norms. ℓ^2 -norm $\|x\|_2 = \sqrt{\sum_i x_i^2}$, ℓ^1 $\|x\|_1 = \sum_i |x_i|$, ℓ^∞ $\|x\|_\infty = \max_i |x_i|$. The ℓ^1 norm promotes sparsity — a fact we will use in Week 3.

Inner product. $\langle x, y \rangle = x^\top y$ in Euclidean space. Norms are induced from inner products via $\|x\| = \sqrt{\langle x, x \rangle}$.

Eigenvector / eigenvalue. $Av = \lambda v$. For symmetric A , eigenvectors are orthogonal and eigenvalues are real; we will exploit this repeatedly (PCA, kPCA, MRF analysis).

Positive (semi)definite matrix. $x^\top A x \geq 0$ for all x . Covariance matrices are PSD. Required as kernel matrices in Week 4.

Random variable, PDF, CDF. If X has PDF p_X , then $\Pr(X \leq a) = \int_{-\infty}^a p_X(t) dt$.

Gaussian distribution. $\mathcal{N}(\mu, \sigma^2)$: $p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$.

Poisson distribution. $\Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$. Mean = variance = λ . Models photon counting.

Bayes’ theorem. $p(x | y) = p(y | x) p(x) / p(y)$. The denominator $p(y)$ does not depend on x and is usually ignored when we *maximise* the posterior over x .

MLE vs. MAP. Maximum-likelihood estimate: $\hat{x}_{\text{MLE}} = \arg \max_x p(y | x)$. Maximum a-posteriori: $\hat{x}_{\text{MAP}} = \arg \max_x p(y | x) p(x)$. MAP = MLE plus a prior.

Gradient descent. $x_{k+1} = x_k - \eta \nabla f(x_k)$, $\eta > 0$ a step size.

2.3 Pointers

3Blue1Brown’s *Essence of Linear Algebra* is the right intuition playlist; StatQuest’s videos on probability vs. likelihood and on Bayes’ theorem are the right intuition for probability. For depth, MIT OCW 18.06 (Strang) and 6.041 (Tsitsiklis) are still the gold standard. The full per-week resource list is in `week_1/Readme.md`.

3 Week 2 — Statistical priors and image denoising

3.1 Big picture

Week 2 is the first time we use (1) for real. The problem is *denoising*: given a noisy image y , recover a clean image $\hat{x}$.

We need two ingredients:

1. A **noise model** $p(y | x)$, that depends on the modality. Gaussian for thermal noise; Poisson for photon-counting (X-ray, CT, microscopy); Rician for the magnitude of an MRI image; Compound-Poisson for polychromatic CT; Rician/Rayleigh/K for ultrasound.
2. A **prior** $p(x)$ that encodes what a “plausible” image looks like — usually a mixture of (i) intensities vary smoothly inside organs and (ii) large jumps are allowed only across organ boundaries.

The right mathematical object for the prior is a **Markov Random Field (MRF)**. An MRF is a probability distribution over an image (treated as a vector of N random variables, one per pixel) in which each pixel is conditionally independent of all distant pixels given its neighbours. By the **Hammersley–Clifford theorem**, every (positive) MRF can be written as a **Gibbs distribution**,

$$p(x) = \frac{1}{Z} \exp\left(-\frac{1}{T} \sum_{c \in C} V_c(x_c)\right), \quad (2)$$

where the sum is over *cliques* c (subsets of pixels that are all pairwise neighbours), V_c are *clique potentials* you get to design, T is a temperature, and Z is the normalising *partition function*.

Different choices of clique potential give different priors. A quadratic potential $V(x_i, x_j) = (x_i - x_j)^2$ gives a **Gaussian MRF**; it strongly penalises any intensity jump and therefore blurs edges. **Edge-preserving** potentials (Huber, generalised Gaussian, truncated quadratic, Mumford–Shah) cap the penalty for large jumps and so preserve organ boundaries. **Ising** (binary labels) and **Potts** (multi-label) priors do the analogous job for segmentation, which we will use in Week 4.

Putting it together, **Bayesian denoising = MAP estimation**:

$$\hat{x} = \arg \max_x p(y | x) p(x) = \arg \min_x \left[\underbrace{-\log p(y | x)}_{\text{data fidelity}} + \underbrace{-\log p(x)}_{\text{regularisation}} \right]. \quad (3)$$

The negative-log Gibbs prior *is* the regulariser you have already seen in machine learning. A Gaussian prior gives ℓ^2 (Tikhonov / ridge) regularisation; a Laplace prior on pixel differences gives *total variation* (TV) (ℓ^1); an MRF with edge-preserving potentials gives a robust regulariser.

3.2 Vocabulary you will meet

Random field. A collection of random variables $\{X_i\}$ indexed by sites i on a grid.

Neighbourhood system. For each site i , a set N_i of neighbouring sites. 4-/8-neighbour in 2D; 6-/18-/26-neighbour in 3D.

Clique. A subset of sites that are all pairwise neighbours (so for a 4-neighbour 2D system, cliques are singletons and edges only).

Markov property. $p(x_i | x_{S \setminus \{i\}}) = p(x_i | x_{N_i})$.

Hammersley–Clifford theorem (1971). Under positivity ($p(x) > 0$), an MRF and a Gibbs distribution on the same neighbourhood system are the same object.

Clique potential V_c . A scalar function over the values of pixels in a clique c ; you design these to encode “smoothness” or “sparsity” or “label agreement”.

Partition function Z . The normaliser in (2). Almost always intractable to compute in closed form; we can still optimise the energy without ever evaluating Z .

Ising / Potts model. Binary / multi-label MRFs whose clique potential is $V(x_i, x_j) = \mathbb{K}[x_i \neq x_j]$.

Gaussian MRF. Quadratic potentials; closed-form posterior; smooths edges.

Edge-preserving prior. Potential that grows slowly (or saturates) for large differences. Huber: $V(t) = t^2/2$ for $|t| \leq \delta$ and $\delta|t| - \delta^2/2$ otherwise.

Gaussian noise. $y_i = x_i + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, i.i.d.

Poisson noise. $y_i \sim \text{Poisson}(\lambda_i)$ with λ_i a function of the underlying intensity x_i and the imaging geometry.

Compound Poisson. Sum of N i.i.d. random variables where N itself is Poisson; arises in polychromatic CT.

Rician distribution. Distribution of the magnitude $\sqrt{a^2 + b^2}$ when $(a, b) \sim \mathcal{N}((\nu \cos \phi, \nu \sin \phi), \sigma^2 I)$. Its PDF involves the modified Bessel function I_0 . Used for MRI magnitude images. Reduces to Rayleigh when $\nu = 0$.

Signal-to-noise ratio. $\text{SNR} = \mathbb{E}[X]/\text{SD}(X)$. For Poisson, $\text{SNR} = \sqrt{\lambda}$.

Total variation (TV). Regulariser $\text{TV}(x) = \sum_i \|\nabla x\|_2$ (isotropic) or $\sum_i (|x_{i+1} - x_i| + |x_{i+\text{row}} - x_i|)$ (anisotropic). Promotes piecewise-constant images.

Non-local means. A denoising algorithm that averages each pixel with all pixels whose *patch* is similar; not strictly MAP, but a beautifully simple algorithm rooted in self-similarity.

3.3 Pointers

Bouman's free PDF *Model-Based Image Processing* is the natural textbook for this week and Week 3. Geman & Geman (1984) and Buades et al. (2005) are the two classical papers to read if you want depth. The Gudbjartsson & Patz (1995) Rician note is a beautiful 4-page read.

4 Week 3 — Inverse problems and the physics of CT and MRI

4.1 Big picture

In Week 2 we recovered $\hat{x}$ from a noisy version of itself. In Week 3 we make it harder: we observe y that has gone through a *forward operator* G that scrambles the image:

$$y = Gx + n. \quad (4)$$

This is called an **inverse problem**. In CT, G is the discrete Radon transform (sum of all line integrals through the patient); in MRI, G is a (typically under-sampled) 2D Fourier transform. In both cases G may be *rank-deficient* (you don't get enough measurements to invert) or *ill-conditioned* (small noise in y blows up in $\hat{x}$). This is what “**ill-posed**” means.

MAP reconstruction fixes ill-posedness by adding a prior:

$$\hat{x} = \arg \min_x \frac{1}{2\sigma^2} \|Gx - y\|_2^2 + \lambda R(x), \quad (5)$$

where $R(x)$ is the negative-log Week-2 prior (Tikhonov, edge-preserving MRF, TV, dictionary, ...) and λ trades data-fidelity against the prior.

To solve (5) you need **multivariate gradient descent** (and friends: Newton, quasi-Newton, ADMM, primal-dual). The gradient of the data term is

$$\nabla_x \frac{1}{2} \|Gx - y\|_2^2 = G^\top (Gx - y). \quad (6)$$

Derive it once by hand using the chain rule with $w := Gx - y$, $f := \frac{1}{2} w^\top w$. For MRI, x is complex-valued, and you need to be careful with what “*gradient of a real-valued function of a complex argument*” means; this is the only place the **Cauchy–Riemann equations** show up in this course.

4.2 CT physics

X-rays are short-wavelength electromagnetic radiation. When an X-ray of intensity I_0 passes through a thin slab of tissue of thickness Δx with **attenuation coefficient** $A(x)$, a fraction $A(x)\Delta x$ of the photons get absorbed. This linear relationship, in the infinitesimal limit, is the **Beer–Lambert law**:

$$I_1 = I_0 \exp\left(-\int_{x_0}^{x_1} A(x) dx\right). \quad (7)$$

A single X-ray therefore gives you a *line integral* of A . One projection is not enough to recover $A(\cdot)$; to do tomography, the scanner takes many such projections from many angles. The collection of all line integrals at all angles is the **Radon transform** of the image.

The **Fourier slice theorem** says that the 1D Fourier transform of a 1D parallel projection at angle θ equals a 1D radial slice through the 2D Fourier transform of the image at the same angle. This is the basis of **filtered back-projection (FBP)** — the direct, non-iterative CT reconstruction algorithm. Iterative **algebraic reconstruction (ART)** and MAP estimation are the alternatives, used when the data are sparse-view, low-dose, or metal-corrupted.

4.3 MRI physics

The human body is mostly water, and water is rich in ^1H hydrogen nuclei (protons). A spinning proton has a magnetic dipole moment; placed in a strong static magnetic field B_0 , the dipoles align and *precess* around B_0 at the **Larmor frequency** $\omega_0 = \gamma B_0$, where γ is the gyromagnetic ratio. For ^1H , $\gamma \approx 42.58$ MHz/T, so at 3 T you have $\omega_0 \approx 128$ MHz.

The dynamics of the aggregate magnetisation $\mathbf{M}(t) = (m_x, m_y, m_z)$ are described by the **Bloch equations**: a system of ODEs containing three terms — precession, longitudinal relaxation toward m_{eq} with time constant T_1 , and transverse relaxation toward 0 with time constant T_2 . T_1 and T_2 depend on tissue, which is why MRI can distinguish soft tissues. A short RF pulse tips the magnetisation into the transverse plane; gradient fields encode spatial position into the precession frequency (*frequency encoding*) and phase (*phase encoding*). Each MRI measurement is one complex sample in **k -space** (the 2D/3D Fourier domain of the image). An *inverse FFT* of a complete k -space dataset gives the image.

When we sub-sample k -space (to image faster), the inverse FFT aliases; the cure is to combine the sparse measurement with a prior that gives a unique solution — the **compressed-sensing MRI** story that you will meet at the end of the week.

4.4 Vocabulary you will meet

Forward operator G . The matrix that maps the unknown image to the measurement. Radon transform for CT; Fourier transform for MRI.

Ill-posed. Either no solution exists, the solution is not unique, or it does not depend continuously on the data (Hadamard’s definition).

Tikhonov regularisation. $R(x) = \|x\|_2^2$ (or $\|Lx\|_2^2$ for some linear L , e.g. a gradient).

Total variation (TV). $R(x) = \|\nabla x\|_1$. The ℓ^1 on gradients makes the prior sparse on edges.

Line search. Pick the step size η in gradient descent by minimising $f(x_k - \eta g_k)$ along the ray. Variants: backtracking, golden-section, Newton’s method.

Matrix calculus. The Jacobian $\partial f / \partial x$ has m rows (outputs) and n columns (inputs); the gradient of a scalar function is the transpose of the Jacobian (so it lives in the same space as x).

Cauchy–Riemann equations. $u_x = v_y$, $u_y = -v_x$ for $f(x + iy) = u(x, y) + iv(x, y)$ to be complex-differentiable. We invoke them only to justify treating complex MRI gradients via the “*complex-as-doubled-real*” trick.

Attenuation coefficient. Local rate of photon absorption per unit length. Units: 1/length.

Hounsfield unit. Affine rescaling of attenuation coefficient with water at 0 HU and air at -1000 HU.

Beer–Lambert law. Eq. (7).

Radon transform. Collection of all parallel line-integrals through an image at all angles.

Fourier slice theorem. Links 1D projections to 2D Fourier slices; the theoretical basis of FBP.

Filtered back-projection (FBP). Closed-form CT reconstruction by applying a ramp filter to each projection, then back-projecting.

ART (algebraic reconstruction). Iteratively project the current estimate onto each measurement constraint; converges to a least-norm solution.

Larmor frequency. $\omega_0 = \gamma B_0$.

Bloch equations. ODEs for the magnetisation vector; contain T_1 and T_2 relaxation times.

T_1 / T_2 . Tissue-dependent relaxation times. T_1 governs longitudinal recovery, T_2 transverse decay.

k -space. The 2D/3D Fourier domain of the image; what the MRI scanner actually samples.

Compressed sensing. Recover a signal from sub-Nyquist samples by exploiting sparsity in some transform domain.

4.5 Pointers

Kak & Slaney’s *Principles of Computerized Tomographic Imaging* and Bouman’s *MBIP* are the two free textbooks to keep at hand. For MRI, watch any one of Allen Ye’s videos and read the iMaio’s k -space explainer; for compressed sensing read the Lustig–Donoho SparseMRI paper.

5 Week 4 — Shape, classical segmentation, registration, kernels

5.1 Big picture

We now switch from *recovering an image* to *recovering structure*: which voxels belong to which tissue (**segmentation**), how a population of anatomical structures varies in geometry (**shape analysis**), how to align two scans to the same coordinate frame (**registration**), and an extra toolkit (**kernel methods**) that everything from SVMs to kernel PCA to many shape models depends on.

5.1.1 Segmentation

The simplest segmenter is **k -means clustering**: partition N feature vectors $\{x_n\}$ into K clusters by minimising

$$\sum_{k=1}^K \sum_{n \in S_k} \|x_n - \mu_k\|_2^2. \quad (8)$$

Solved by alternating optimisation: fix means, update assignments (each point goes to nearest mean); fix assignments, update means (set each μ_k to the centroid of its cluster). It always converges (to a local minimum); finding the global minimum is NP-hard. **k -means++** is a clever initialisation that gives a provable approximation factor. **k -medoids** replaces means by data points and gives outlier robustness. The Voronoi cells of the means partition feature space; if you switch the distance from Euclidean to Manhattan, you switch from straight-edge to axis-aligned cells.

Gaussian Mixture Models (GMMs) are the soft-assignment generalisation: each cluster is a Gaussian, points have a *posterior probability* of belonging to each cluster, and the parameters

are estimated by the **Expectation–Maximisation (EM)** algorithm. EM is a coordinate-ascent algorithm on a lower bound of the log-likelihood (the same machinery you will reuse in Week 6 for the VAE’s ELBO).

Graph cuts formulate segmentation as min-cut / max-flow on a graph whose edges encode unary (label likelihood) and pairwise (smoothness) potentials — in other words, MAP inference on an MRF with binary or multi-label cliques. **Spectral clustering** uses the eigenvectors of the graph Laplacian instead. **Level sets** (Chan–Vese, Mumford–Shah) evolve an implicit curve by a PDE whose driving force is an energy.

5.1.2 Shape analysis

A **shape** is what remains of an outline after you remove translation, rotation, and isotropic scale — formally, an equivalence class under the **similarity group**. We represent a shape by an ordered **pointset** of landmarks placed consistently across all subjects; that fixes correspondences and lets us write each shape as a vector in $\mathbb{R}^{DN}$.

To remove the nuisance transformations we use **Procrustes analysis**: translate to the origin, scale to unit norm, and rotate to minimise the sum-of-squared distances to a reference shape (closed form via SVD of the cross-covariance). **Generalised Procrustes Analysis (GPA)** aligns all shapes to a moving mean iteratively. After alignment, the shapes live near a single point on a high-dimensional sphere, and a **tangent-space approximation** lets us run linear PCA on them. The principal components are the **modes of shape variation**; walking $\pm 2\sigma$ along each mode is the classic visualisation. The model that does all of this is the **Active Shape Model (ASM)**; if you add per-vertex intensity information you get the **Active Appearance Model (AAM)**.

5.1.3 Registration

Given two images $R(x)$ and $T(x)$, find a transformation $\phi : \Omega \rightarrow \Omega$ that “aligns” $T \circ \phi$ to R . Transformations are usually rigid, similarity, affine (linear), B-spline (parametric non-linear), or **diffeomorphic** (smooth and invertible, the gold standard for medical work). Similarity measures depend on modality: **sum-of-squared differences (SSD)** for same modality, **normalised cross-correlation (NCC)** for robustness to linear intensity changes, **mutual information (MI)** for *different* modalities (CT $\leftrightarrow$ MRI). Atlas building (e.g. Talairach for the brain) iteratively registers all subjects to a moving mean (same recipe as GPA, but in image space).

5.1.4 Kernels (supplementary)

Many of the methods above (and a few in Week 5) are special cases of a unified picture. Map data nonlinearly through some $\phi : \mathbb{R}^d \rightarrow \mathcal{F}$ into a (possibly infinite-dimensional) Hilbert space $\mathcal{F}$ and do linear methods there. The **kernel trick** is that you never need to construct ϕ explicitly — you just need the inner product $k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathcal{F}}$. The classic example is the **RBF (Gaussian) kernel** $k(x, y) = \exp(-\gamma \|x - y\|^2)$, whose feature space is infinite-dimensional.

A kernel is admissible iff it is a positive-definite function (**Mercer / Moore–Aronszajn**); the corresponding feature space is a **Reproducing Kernel Hilbert Space (RKHS)**. **Kernel PCA** runs PCA in the RKHS using only the **Gram matrix** $K_{ij} = k(x_i, x_j)$. **Support Vector Machines (SVMs)** find a maximum-margin hyperplane in the RKHS; deriving them goes through the **KKT conditions** of constrained convex optimisation.

5.2 Vocabulary you will meet

Cluster compactness. Within a cluster, $\sum_n \|x_n - c\|^2 = \frac{1}{N} \sum_{n,j} \|x_n - x_j\|^2$; the identity that makes k -means about pairwise distances as much as about centroids.

Voronoi diagram. Partition of space by “nearest centre”; the cell shapes depend on the distance.

Expectation–Maximisation (EM). Alternating maximisation of $\mathbb{E}_{q(z)}[\log p(x, z; \theta)] - \text{KL}(q||p(z|x; \theta))$ over q then θ . Same idea reappears in the VAE in Week 6.

Graph Laplacian. $L = D - W$, where D is the degree matrix and W the affinity matrix. Eigenvectors give spectral embeddings.

Min-cut / max-flow. Polynomial-time algorithms for graph cuts in undirected graphs; the workhorse of binary graph-cut segmentation.

Level set. An evolving curve represented as the zero level set of an embedding function ϕ , governed by a PDE $\partial\phi/\partial t = F||\nabla\phi||$.

Procrustes alignment. Closed-form removal of translation, scale, rotation between two pointsets via SVD.

Tangent space. Local linear approximation of a curved manifold (here, shape space).

Active Shape Model. Mean shape $\bar{x}$ plus principal directions $\{v_k\}$: $x \approx \bar{x} + \sum_k b_k v_k$.

Mutual information. $\text{MI}(X; Y) = \text{KL}(p_{XY}||p_X p_Y)$. The right similarity measure for multi-modal registration.

Diffeomorphism. Smooth invertible map with smooth inverse; the gold standard for non-linear medical-image registration.

Kernel trick. Inner products in feature space via $k(x, y)$, never building ϕ .

Positive-definite function. $\sum_{ij} c_i c_j k(x_i, x_j) \geq 0$ for all choices of $\{x_i\}$, $\{c_i\}$.

RKHS / Moore–Aronszajn. Every PD function k is the reproducing kernel of a unique Hilbert space of functions; that’s where $\phi(x) := k(x, \cdot)$ lives.

Gram matrix. $K_{ij} = k(x_i, x_j)$; what kernel PCA actually eigendecomposes.

KKT conditions. First-order optimality conditions for constrained convex programs; derived in Boyd & Vandenberghe ch. 5. You meet them when you derive the SVM dual.

5.3 Pointers

Bishop’s *Pattern Recognition and Machine Learning* chapters 9 (clustering, EM) and 12 (PCA) cover most of this. The von Luxburg spectral-clustering tutorial is the canonical read. Cootes et al. (1995) is the original ASM paper. For kernels and SVMs, Schölkopf & Smola’s *Learning with Kernels* and Boyd & Vandenberghe’s *Convex Optimization* (KKT in ch. 5) are the references.

6 Week 5 — CNNs for medical images

6.1 Big picture

A **convolutional neural network (CNN)** is a stack of layers each of which slides one or more small filters (“kernels”) across its input, applies a non-linearity, possibly downsamples, and feeds the result to the next layer. Two properties make CNNs the right inductive bias for images: **translation equivariance** (same filter, applied everywhere) and **locality** (each output pixel depends only on a neighbourhood of the input).

For **classification**, the typical recipe is $\text{Conv} \rightarrow \text{Norm} \rightarrow \text{ReLU} \rightarrow \text{Pool}$ repeated L times, then a fully-connected head producing class scores. For **segmentation** the classical choice in medical imaging is the **U-Net** (Ronneberger et al., MICCAI 2015): an encoder-decoder with skip connections that concatenate encoder feature maps to the matching-resolution decoder layer. The skips let the decoder recover fine detail that pooling has destroyed.

The **engineering recipe** that wins most medical segmentation challenges is **nnU-Net** (Isensee et al., 2021) — a U-Net plus a careful, dataset-adaptive pipeline (preprocessing, augmentation, training schedule, postprocessing). The take-away is that in medical imaging, *good engineering of a standard architecture often beats a novel architecture*.

Training a CNN means solving a non-convex optimisation problem with **stochastic gradient descent** (and friends: Adam, AdamW). Gradients are computed by **reverse-mode automatic differentiation** (the “backpropagation” algorithm). The whole thing is one giant chain rule on the computational graph.

6.2 Vocabulary you will meet

Convolution layer. Slides a small kernel across the input; each kernel is shared across spatial positions (translation equivariance). Has $C_{\text{in}} \cdot C_{\text{out}} \cdot k \cdot k$ parameters for a $k \times k$ 2D kernel.

Cross-correlation vs. convolution. DL libraries compute cross-correlation; for learning that’s identical to true convolution because the kernel is also learnt.

Stride / padding / dilation. Three knobs that determine output size and receptive field.

Receptive field. Spatial extent of the input that a given output pixel depends on. Grows with depth and dilation.

Activation. Pointwise non-linearity: ReLU $\max(0, x)$; GELU $x \cdot \Phi(x)$ (Gaussian CDF); sigmoid $1/(1 + e^{-x})$; softmax (vector $\rightarrow$ probability simplex); softplus $\log(1 + e^x)$.

Pooling. Downsamples by max or average; mostly replaced by strided convolution in modern nets.

Normalisation. BatchNorm normalises per-channel across a batch; GroupNorm (preferred for small medical batches) per group of channels; InstanceNorm per sample.

Dice loss. $1 - \frac{2|P \cap T|}{|P| + |T|}$. Handles class imbalance better than cross-entropy for medical segmentation.

Backpropagation. Apply the chain rule layer-by-layer in reverse to compute all parameter gradients in one pass.

Adam / AdamW. Adaptive-learning-rate optimisers; *AdamW* (decoupled weight decay) is the default for modern training.

U-Net. Encoder-decoder CNN with skip connections, the medical-segmentation workhorse.

Dice / IoU / Hausdorff 95 / ASSD. Segmentation metrics. *Dice* is overlap; *IoU* is intersection over union; *HD95* is the 95th-percentile Hausdorff distance; *ASSD* is average symmetric surface distance.

6.3 Pointers

Stanford CS231n class notes are the right text. 3Blue1Brown’s first two NN videos for intuition. The original U-Net and nnU-Net papers are short and worth reading in full. The MONAI tutorials are the fastest path to a working medical-imaging PyTorch pipeline.

7 Week 6 — VAEs and GANs

7.1 Big picture

A **generative model** is a probability distribution $p_{\theta}(x)$ over the data x that you can both *score* and (usually) *sample* from. Once you have a generative model, you have a *learned prior* that you can plug back into the Bayesian reconstruction objective (5) from Week 3.

Two classical deep generative families:

7.1.1 Variational Autoencoder (VAE)

Assume x is generated from a latent code z through a decoder $p_\theta(x | z)$, with a simple prior $p(z) = \mathcal{N}(0, I)$. The marginal $p_\theta(x) = \int p_\theta(x | z)p(z)dz$ is intractable. Introduce an **encoder** $q_\phi(z | x) \approx p_\theta(z | x)$, and derive the **Evidence Lower Bound (ELBO)**:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p(z)). \quad (9)$$

Maximise the ELBO over both θ and ϕ jointly with SGD. The first term is a *reconstruction* loss; the second is a *regulariser* pulling the posterior toward the prior. The only trick to make this trainable is the **reparameterisation trick**: write $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ so that the stochasticity is isolated in ϵ and gradients flow through μ and σ .

For two Gaussians $\mathcal{N}(\mu, \sigma^2 I)$ and $\mathcal{N}(0, I)$ in $\mathbb{R}^d$,

$$\text{KL}(\mathcal{N}(\mu, \sigma^2 I) \| \mathcal{N}(0, I)) = \frac{1}{2} \sum_{i=1}^d (\sigma_i^2 + \mu_i^2 - 1 - \log \sigma_i^2). \quad (10)$$

Memorise this; you will reuse it in every VAE you ever write.

7.1.2 Generative Adversarial Network (GAN)

A *generator* $G_\theta : \mathbb{R}^d \rightarrow \mathcal{X}$ maps noise $z \sim p_z$ to samples $G(z)$; a *discriminator* $D_\psi : \mathcal{X} \rightarrow (0, 1)$ tries to tell real samples from generated ones. The **min-max** game is

$$\min_{\theta} \max_{\psi} \mathbb{E}_{x \sim p_{\text{data}}} \log D_\psi(x) + \mathbb{E}_{z \sim p_z} \log(1 - D_\psi(G_\theta(z))). \quad (11)$$

At the optimum (with infinite capacity), D recovers the density ratio and G produces samples from p_{data} . In practice, training is unstable; the standard fixes are **WGAN-GP** (Wasserstein objective + gradient penalty), **spectral normalisation** of D , **TTUR** (separate learning rates), and the generator EMA. Conditional variants (pix2pix, CycleGAN) condition on a second image and are the GAN of choice for medical cross-modality synthesis.

7.1.3 Diffusion models (in one paragraph)

A diffusion model is a Markov chain that progressively adds Gaussian noise to data over T steps; a neural network learns the *reverse* step (the denoiser). Trained well, the reverse chain samples from p_{data} . They have largely replaced GANs for image synthesis. Their use as a *prior* for medical-image reconstruction (e.g. Chung & Ye 2022) is currently one of the hottest research areas.

7.1.4 The thread tying it all together

Look back at (5). Anywhere $R(x)$ appears, you can plug in:

- a hand-designed MRF energy (Week 2);
- the negative log-density of a VAE;
- the score (gradient of log-density) of a diffusion model.

The mathematical scaffolding of weeks 2 and 3 *is* the same one that frames the most modern reconstruction methods of 2024–26.

7.2 Vocabulary you will meet

Latent variable. An unobserved random variable z that *generates* the observation x .

ELBO. Eq. (9). The thing you maximise.

Reparameterisation. Trick to backprop through $z \sim q_\phi(z|x)$ by writing $z = g_\phi(x, \epsilon)$.

Posterior collapse. A failure mode of VAEs where the decoder ignores z and the encoder outputs the prior.

β -VAE. Multiply the KL term by β to trade reconstruction for disentanglement.

Mode collapse. GAN failure where G produces only one (or a few) outputs.

WGAN-GP. Wasserstein loss with a gradient penalty for stable adversarial training.

Spectral normalisation. Constrain the spectral norm of D 's weights to enforce Lipschitzness.

Pix2pix / CycleGAN. Paired and unpaired image-to-image translation GANs.

DDPM. Denoising Diffusion Probabilistic Model (Ho, Jain & Abbeel 2020).

Score function. $\nabla_x \log p(x)$; what a diffusion / score-based model approximates.

7.3 Pointers

Lilian Weng's blog is the best single source of explainers for VAEs and GANs. The original Kingma–Welling (VAE) and Goodfellow (GAN) papers are short and worth reading in full. For diffusion, Lilian Weng's blog again, plus the DDPM paper. For the medical application of generative priors, Chung & Ye (2022) is the cleanest read.

8 Week 7 — Paper reading + guided implementation

8.1 Big picture

You now have the vocabulary of weeks 1–6. The point of week 7 is to *prove it* by reading a real research paper and reproducing one experiment from it. The strongest signal that you understand a method is not that you can summarise it — it's that you can rebuild it.

8.1.1 Reading method

Use Andrew Ng's three-pass method:

1. Pass 1 (5–10 min): title, abstract, figures, conclusions. Decide whether to keep going.
2. Pass 2 (~1 h): read carefully but skip proofs. Write a one-page summary covering: problem, prior work and its limitation, key idea, method, experiments, results, limitations.
3. Pass 3 (a day): re-implement the key equation or one experiment. Where you get stuck is exactly where you don't understand it yet.

8.1.2 Choosing a paper

Pick something where:

- the problem connects to weeks 2–6 (don't go to a totally new sub-field this week);
- there is *public code* and a *public dataset*;
- the paper is *at most 12 pages*;
- the venue is MICCAI, MedIA, TMI, IPMI, MIDL, NeurIPS.

A starter pool is in `week_7/Readme.md`; Prof. Awate's `Slides_ProjectTopicExamples.pdf` is another good source.

8.2 Deliverables

A one-page summary, hand-written derivations of the central equations, a reproduction notebook, and a short note describing the gap between the paper's claim and your reproduction.

9 Week 8 — Final project

9.1 Deliverables

A 6–10 page report, 10–15 talk slides, a clean reproducible `code/` folder (with `requirements.txt`), a `results/` folder, and a short live demo.

9.2 Grading dimensions (mirroring CS-736)

- Difficulty of the chosen topic.
- Depth of theoretical reading and understanding.
- Depth of empirical analysis: have you compared against a baseline? Have you run on a real test set with appropriate metrics?
- Quality of the slides.
- Reproducibility of the code.

9.3 Suggested project shapes

- Bayesian denoising with a learned prior — CNN denoiser vs. MRF MAP baseline.
- Under-sampled MRI reconstruction — model-based unrolled network (e.g. MoDL) on fastMRI vs. zero-filled baseline.
- Brain-tumour segmentation — U-Net on BraTS vs. nnU-Net out-of-the-box.
- Shape-prior segmentation — ASM (week 4) combined with a CNN initialiser.
- Cross-modality synthesis — CycleGAN MR $\leftrightarrow$ CT on a small paired dataset.
- VAE anomaly detection on chest X-rays.

9.4 Common mistakes to avoid

No baseline. Training loss as the headline metric. Tiny test set with no confidence interval. No qualitative failure analysis. Code that won't run on someone else's machine.

Closing note

You have, over eight weeks, walked from “what is a pixel” to “score-based diffusion priors for MRI”. The vocabulary you have now is enough to read the proceedings of MICCAI cover-to-cover and to join, contribute to, or lead, a medical-imaging research project.

Good luck. The work in this field is meaningful; many of the people who do it make patients' lives demonstrably better. Take that seriously.